

Physics 513 QFT HW 6

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1. 1

The interaction Lagrangian is

$$\mathcal{L}_{\text{int}} = \frac{G}{3!}\Phi^3 + \frac{C}{4!}\Phi^4, \quad (1)$$

where I use Φ for interacting fields and ϕ for free fields. I have re-normalized the coupling constants, following Schwartz, so that the math works out easier. To restore the original couplings just use $G = -3!g = -6g$ and $C = -4!\lambda = -24\lambda$. I use the notation I developed in my Real Scalar Field Notes.

Each 3-interaction vertex has 3 lines, and gets a factor of iG . Each 4-interaction vertex has 4 lines, and gets a factor of iC .

The matrix element $\mathcal{M}$ is defined by

$$\delta^{(4)}(\vec{P}_i - \vec{P}_f) i\mathcal{M} = \mathfrak{G}[s] + \mathfrak{G}[t] + \mathfrak{G}[u] + \mathfrak{G}[4], \quad (2)$$

where $\vec{P}_i = \vec{p}_1 + \vec{p}_2$ is the total incoming momentum and $\vec{P}_f = \vec{p}_3 + \vec{p}_4$ is the total outgoing momentum. Figures 1-4 show the diagrams for the $\mathfrak{G}[i]$. Since this is 2 to 2 scattering I can simplify my answer with Mandelstam variables

$$\begin{aligned} s &= (\vec{p}_1 + \vec{p}_2)^2 = (\vec{p}_3 + \vec{p}_4)^2, \\ t &= (\vec{p}_1 - \vec{p}_3)^2 = (\vec{p}_2 - \vec{p}_4)^2, \\ u &= (\vec{p}_1 - \vec{p}_4)^2 = (\vec{p}_2 - \vec{p}_3)^2. \end{aligned} \quad (3)$$

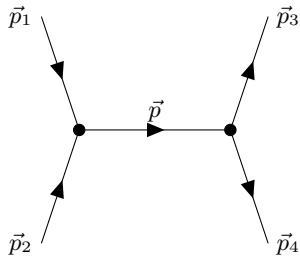


FIG. 1. Feynman diagram for $\mathfrak{G}[s]$.

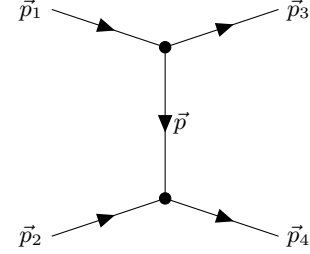


FIG. 2. Feynman diagram for $\mathfrak{G}[t]$.

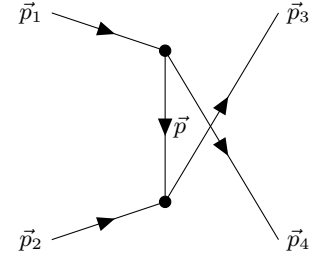


FIG. 3. Feynman diagram for $\mathfrak{G}[u]$. Notice that the crossing between $\vec{p}_3$ and $\vec{p}_4$ does not have a dot. That means it is not a vertex.

$$\mathfrak{G}[s] = i(iG)^2 \int d^4p \frac{\delta^{(4)}(\vec{p}_1 + \vec{p}_2 - \vec{p}) \delta^{(4)}(\vec{p} - \vec{p}_3 - \vec{p}_4)}{\vec{p}^2 - m^2 + i\epsilon} \quad (4)$$

$$\mathfrak{G}[s] = -iG^2 \frac{\delta^{(4)}(\vec{p}_1 + \vec{p}_2 - \vec{p}_3 - \vec{p}_4)}{(\vec{p}_1 + \vec{p}_2)^2 - m^2 + i\epsilon} \quad (5)$$

$$\mathfrak{G}[s] = -iG^2 \frac{\delta^{(4)}(\vec{p}_1 + \vec{p}_2 - \vec{p}_3 - \vec{p}_4)}{s - m^2} \quad (6)$$

$$\mathfrak{G}[s] = -iG^2 \frac{\delta^{(4)}(\vec{P}_i - \vec{P}_f)}{s - m^2} \quad (7)$$

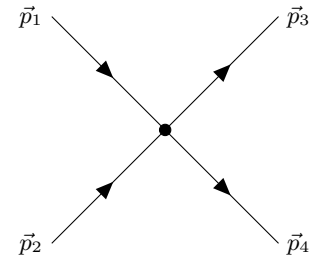


FIG. 4. Feynman diagram for $\mathfrak{G}[4]$.

$$\mathfrak{V}[t] = i(iG)^2 \int d^4p \frac{\delta^{(4)}(\vec{p}_1 - \vec{p}_3 - \vec{p})\delta^{(4)}(\vec{p}_2 + \vec{p} - \vec{p}_4)}{\vec{p}^2 - m^2 + i\epsilon} \quad (8)$$

$$\mathfrak{V}[t] = -iG^2 \frac{\delta^{(4)}(\vec{P}_i - \vec{P}_f)}{t - m^2} \quad (9)$$

$$\mathfrak{V}[u] = i(iG)^2 \int d^4p \frac{\delta^{(4)}(\vec{p}_1 - \vec{p}_4 - \vec{p})\delta^{(4)}(\vec{p}_2 + \vec{p} - \vec{p}_3)}{\vec{p}^2 - m^2 + i\epsilon} \quad (10)$$

$$\mathfrak{V}[u] = -iG^2 \frac{\delta^{(4)}(\vec{P}_i - \vec{P}_f)}{u - m^2} \quad (11)$$

$$\mathfrak{V}[4] = iC\delta^{(4)}(\vec{p}_1 + \vec{p}_2 - \vec{p}_3 - \vec{p}_4) = iC\delta^{(4)}(\vec{P}_i - \vec{P}_f) \quad (12)$$

$$\cancel{\delta^{(4)}(\vec{P}_i - \vec{P}_f)} i\mathcal{M} = i\cancel{\delta^{(4)}(\vec{P}_i - \vec{P}_f)} \left[C - G^2 \left(\frac{1}{s - m^2} + \frac{1}{t - m^2} + \frac{1}{u - m^2} \right) \right] \quad (13)$$

$$\mathcal{M} = C - G^2 \left(\frac{1}{s - m^2} + \frac{1}{t - m^2} + \frac{1}{u - m^2} \right) \quad (14)$$

$$\boxed{\mathcal{M} = 36g^2 \left(\frac{1}{s - m^2} + \frac{1}{t - m^2} + \frac{1}{u - m^2} \right) - 24\lambda} \quad (15)$$

2. 2

2.1. Lagrangian

$$2\mathcal{L} = (\partial\phi)^2 - m^2\phi^2 \quad (16)$$

I introduce the shorthand

$$\eta^{\mu\nu}(\partial_\mu\phi)(\partial_\nu\phi) = (\partial\phi)^2. \quad (17)$$

$$\Phi = \phi + \lambda\phi^2 \quad (18)$$

$$\phi = \frac{-1 \pm \sqrt{1 + 4\lambda\Phi}}{2\lambda} \quad (19)$$

Which sign should I take? It seems that I should insist that $\lambda \rightarrow 0 \implies \Phi \rightarrow \phi$. Only the + solution satisfies this condition. In fact the - diverges.

$$\frac{-1 - \sqrt{1 + 4\lambda\Phi}}{2\lambda} = -\frac{1}{\lambda} - \Phi + \mathcal{O}(\lambda) \quad (20)$$

$$\phi = \frac{-1 + \sqrt{1 + 4\lambda\Phi}}{2\lambda} = \Phi - \lambda\Phi^2 + 2\lambda^2\Phi^3 + \mathcal{O}(\lambda^3) \quad (21)$$

So I take the + solution. I assume λ is small and I am only looking at the leading perturbative correction. I only need to expand ϕ to order λ^2 to get that correction.

$$\phi^2 = (\Phi - \lambda\Phi^2 + 2\lambda^2\Phi^3)^2 = \Phi^2 - 2\lambda\Phi^3 + 5\lambda^2\Phi^4 + \mathcal{O}(\lambda^3) \quad (22)$$

$$\partial_\alpha\phi = \partial_\alpha\Phi - 2\lambda\Phi\partial_\alpha\Phi + 6\lambda^2\Phi^2\partial_\alpha\Phi \quad (23)$$

$$(\partial\phi)^2 = \eta^{\alpha\beta}(\partial_\alpha\phi)(\partial_\beta\phi) \quad (24)$$

I only keep terms up to order λ^2 .

$$\begin{aligned} (\partial\phi)^2 &= \eta^{\alpha\beta}(\partial_\alpha\Phi - 2\lambda\Phi\partial_\alpha\Phi + 6\lambda^2\Phi^2\partial_\alpha\Phi) \\ (\partial_\beta\Phi - 2\lambda\Phi\partial_\beta\Phi + 6\lambda^2\Phi^2\partial_\beta\Phi) &= (\partial\Phi)^2 - 4\lambda\Phi(\partial\Phi)^2 \\ &\quad + 16\lambda^2\Phi^2(\partial\Phi)^2 \end{aligned} \quad (25)$$

$$(\partial\phi)^2 = (\partial\Phi)^2 - 4\lambda\Phi(\partial\Phi)^2 + 16\lambda^2\Phi^2(\partial\Phi)^2 \quad (26)$$

$$2\mathcal{L}_0 = (\partial\Phi)^2 - m^2\Phi^2 \quad (27)$$

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_0 + \lambda m^2\Phi^3 - 2\lambda\Phi(\partial\Phi)^2 + 8\lambda^2\Phi^2(\partial\Phi)^2 \\ &\quad - \frac{5}{2}\lambda^2 m^2\Phi^4 + \mathcal{O}(\lambda^3) \end{aligned} \quad (28)$$

I simplify the Lagrangian using integration by parts.

$$-2\lambda\Phi(\partial\Phi)^2 = -2\lambda\eta^{\alpha\beta}\Phi(\partial_\alpha\Phi)(\partial_\beta\Phi) \quad (29)$$

$$\partial_\alpha\Phi^2 = 2\Phi\partial_\alpha\Phi \quad (30)$$

$$-2\lambda\Phi(\partial\Phi)^2 = -\lambda\eta^{\alpha\beta}(\partial_\alpha\Phi^2)(\partial_\beta\Phi) = \lambda\Phi^2\eta^{\alpha\beta}\partial_\alpha\partial_\beta\Phi \quad (31)$$

$$\eta^{\alpha\beta}\partial_\alpha\partial_\beta = \partial_t^2 - \partial_x^2 - \partial_y^2 - \partial_z^2 = \partial_t^2 - \nabla^2 = \square \quad (32)$$

$$-2\lambda\Phi(\partial\Phi)^2 = \lambda\Phi^2\square\Phi \quad (33)$$

$$8\lambda^2\Phi^2(\partial\Phi)^2 = 8\lambda^2\Phi^2\eta^{\alpha\beta}(\partial_\alpha\Phi)(\partial_\beta\Phi) = \frac{8}{3}\lambda^2\eta^{\alpha\beta}(\partial_\alpha\Phi^3)(\partial_\beta\Phi) = \quad (34)$$

$$\partial_\alpha\Phi^3 = 3\Phi^2\partial_\alpha\Phi \quad (35)$$

$$\begin{aligned} 8\lambda^2\Phi^2(\partial\Phi)^2 &= \frac{8}{3}\lambda^2\eta^{\alpha\beta}(\partial_\alpha\Phi^3)(\partial_\beta\Phi) = \\ &= -\frac{8}{3}\lambda^2\Phi^3\eta^{\alpha\beta}\partial_\alpha\partial_\beta\Phi = -\frac{8}{3}\lambda^2\Phi^3\square\Phi \end{aligned} \quad (36)$$

$$\boxed{\mathcal{L} = \mathcal{L}_0 + \lambda m^2\Phi^3 + \lambda\Phi^2\square\Phi - \frac{8}{3}\lambda^2\Phi^3\square\Phi - \frac{5}{2}\lambda^2 m^2\Phi^4} \quad (37)$$

2.2. Feynman Rules

I give the Feynman rules for terms in the interaction Lagrangian. I give the factor each interaction term contributes to a vertex. I do this by putting the interaction terms into the standard form used in Schwartz and my Real Scalar Field Notes.

$$\lambda m^2 \Phi^3 = \frac{3! \lambda}{3!} m^2 \Phi^3 = \frac{6\lambda}{3!} m^2 \Phi^3 \quad (38)$$

$$\boxed{\lambda m^2 \Phi^3 \Leftrightarrow 6i\lambda m^2} \quad (39)$$

I denote this 3-way vertex with a circle. The other 3-way vertex is

$$\lambda \Phi^2 \square \Phi = \frac{2! \lambda}{2!} \Phi^2 \square \Phi = \frac{2\lambda}{2!} \Phi^2 \square \Phi \quad (40)$$

where the $2!$ comes from the 2 indistinguishable scalar bosons in Φ^2 . If momenta $\vec{p}_1$, $\vec{p}_2$, and $\vec{p}_3$ meet at this vertex I need to account for each one being the $\square \Phi$ boson, so the Feynman rule is

$$\boxed{\lambda \Phi^2 \square \Phi \Leftrightarrow -2i\lambda[(\vec{p}_1)^2 + (\vec{p}_2)^2 + (\vec{p}_3)^2]} \quad (41)$$

where each $\square \Phi$ contributes $-\vec{p}(\square \Phi)^2$. I denote this 3-way vertex with a square.

$$-\frac{5}{2} \lambda^2 m^2 \Phi^4 = \frac{-60 \lambda^2 m^2}{4!} \Phi^4 \quad (42)$$

$$\boxed{-\frac{5}{2} \lambda^2 m^2 \Phi^4 \Leftrightarrow -60i\lambda^2 m^2} \quad (43)$$

I denote this 4-way with a circle.

$$-\frac{8}{3} \lambda^2 \Phi^3 \square \Phi = \frac{-16 \lambda^2}{3!} \Phi^3 \square \Phi \quad (44)$$

Let the momenta at this 4-way be $\vec{p}_1$, $\vec{p}_2$, $\vec{p}_3$, and $\vec{p}_4$.

$$\boxed{-\frac{8}{3} \lambda^2 \Phi^3 \square \Phi \Leftrightarrow 16i\lambda^2[(\vec{p}_1)^2 + (\vec{p}_2)^2 + (\vec{p}_3)^2 + (\vec{p}_4)^2]} \quad (45)$$

I denote this 4-way with a square.

2.3. Matrix Element

I calculate the matrix element $\mathcal{M}$ to leading order for $\Phi\Phi \rightarrow \Phi\Phi$ scattering. The initial and final momenta respectively are

$$\vec{P} = \vec{p}_1 + \vec{p}_2, \quad \vec{K} = \vec{p}_3 + \vec{p}_4. \quad (46)$$

Because this is 2 to 2 scattering I use the Mandelstam variables in Eq. 3. The matrix element is

$$\delta^{(4)}(\vec{P} - \vec{K}) i\mathcal{M} = \mathfrak{Y}[1] + \mathfrak{Y}[2] + \sum_{n=1}^{12} \mathfrak{Y}[n] \quad (47)$$

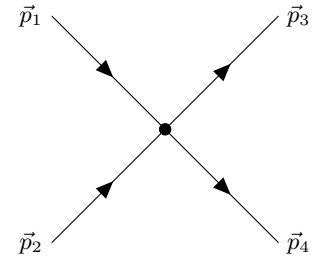


FIG. 5. Feynman diagram $\mathfrak{Y}[1]$.

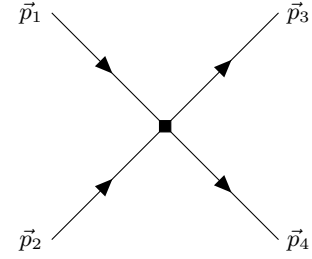


FIG. 6. $\mathfrak{Y}[2]$

where the $\mathfrak{Y}$ are 4-way diagrams and the $\mathfrak{Y}$ are 3-way diagrams. I show the Feynman diagrams in Figs. 5-18.

$$\begin{aligned} \mathfrak{Y}[1] &= -60i\lambda^2 m^2 \delta^{(4)}(\vec{p}_1 + \vec{p}_2 - \vec{p}_3 - \vec{p}_4) = \\ &= -60i\lambda^2 m^2 \delta^{(4)}(\vec{P} - \vec{K}) \end{aligned} \quad (48)$$

$$\mathfrak{Y}[2] = 16i\lambda^2[(\vec{p}_1)^2 + (\vec{p}_2)^2 + (\vec{p}_3)^2 + (\vec{p}_4)^2] \delta^{(4)}(\vec{P} - \vec{K}) \quad (49)$$

For any on-shell scalar boson Φ : $\vec{p}^2 = m^2$.

$$\mathfrak{Y}[2] = 64i\lambda^2 m^2 \delta^{(4)}(\vec{P} - \vec{K}) \quad (50)$$

$$\begin{aligned} \mathfrak{Y}[1] &= (6i\lambda m^2)^2 i \int d^4 p \frac{\delta^{(4)}(\vec{P} - \vec{p}) \delta^{(4)}(\vec{p} - \vec{K})}{\vec{p}^2 - m^2 + i\epsilon} = \\ &= -(6\lambda m^2)^2 i \frac{\delta^{(4)}(\vec{P} - \vec{K})}{s - m^2} \end{aligned} \quad (51)$$

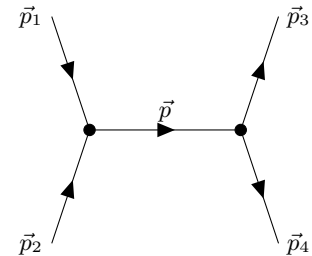
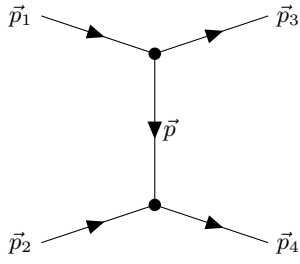
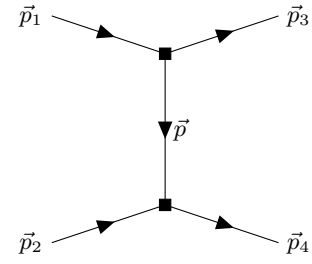
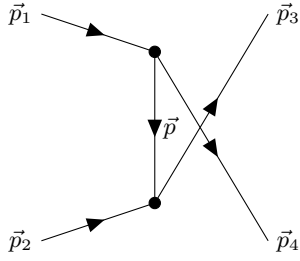
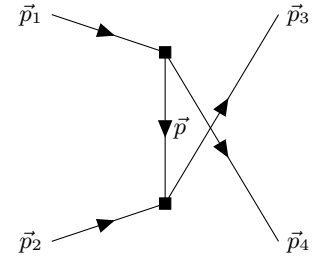
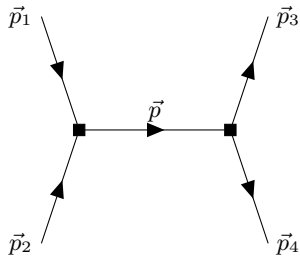


FIG. 7. $\mathfrak{Y}[1]$

FIG. 8. $\mathfrak{V}[2]$ FIG. 11. $\mathfrak{V}[5]$ FIG. 9. $\mathfrak{V}[3]$ FIG. 12. $\mathfrak{V}[6]$

$$\begin{aligned} \mathfrak{V}[2] &= -(6\lambda m^2)^2 i \\ &\int d^4 p \frac{\delta^{(4)}(\vec{p}_1 - \vec{p}_3 - \vec{p}) \delta^{(4)}(\vec{p} + \vec{p}_2 - \vec{p}_4)}{\vec{p}^2 - m^2 + i\epsilon} = \\ &\quad - (6\lambda m^2)^2 i \frac{\delta^{(4)}(\vec{P} - \vec{K})}{t - m^2} \end{aligned} \quad (52)$$

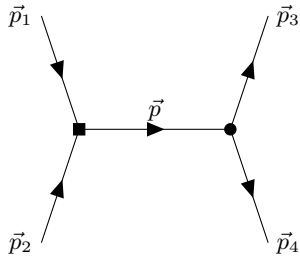
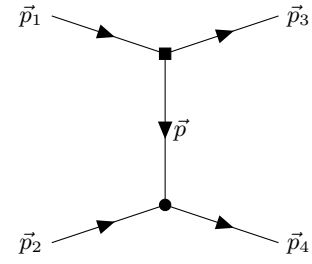
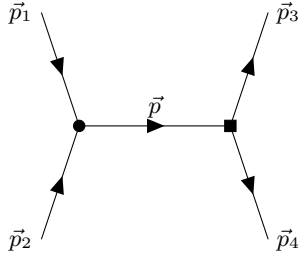
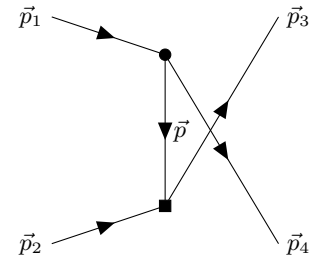
$$\begin{aligned} \mathfrak{V}[3] &= (6i\lambda m^2)^2 i \\ &\int d^4 p \frac{\delta^{(4)}(\vec{p}_1 - \vec{p}_4 - \vec{p}) \delta^{(4)}(\vec{p} + \vec{p}_2 - \vec{p}_3)}{\vec{p}^2 - m^2 + i\epsilon} = \\ &\quad - (6\lambda m^2)^2 i \frac{\delta^{(4)}(\vec{P} - \vec{K})}{u - m^2} \end{aligned} \quad (53)$$

FIG. 10. $\mathfrak{V}[4]$

$$\begin{aligned} \mathfrak{V}[4] &= (-2i\lambda)^2 i \int d^4 p \frac{\delta^{(4)}(\vec{P} - \vec{p}) \delta^{(4)}(\vec{p} - \vec{K})}{\vec{p}^2 - m^2 + i\epsilon} \\ &[(\vec{p}_1)^2 + (\vec{p}_2)^2 + \vec{p}^2][(\vec{p}_3)^2 + (\vec{p}_4)^2 + \vec{p}^2] = \\ &\quad - 4\lambda^2 i \frac{\delta^{(4)}(\vec{P} - \vec{K})}{\vec{P}^2 - m^2} (2m^2 + \vec{P}^2)^2 = \\ &\quad - 4\lambda^2 i \frac{(s + 2m^2)^2}{s - m^2} \delta^{(4)}(\vec{P} - \vec{K}) \end{aligned} \quad (54)$$

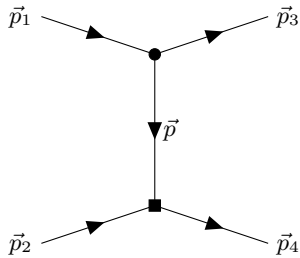
$$\begin{aligned} \mathfrak{V}[5] &= (-2i\lambda)^2 i \int d^4 p \frac{\delta^{(4)}(\vec{p}_1 - \vec{p}_3 - \vec{p}) \delta^{(4)}(\vec{p} + \vec{p}_2 - \vec{p}_4)}{\vec{p}^2 - m^2 + i\epsilon} \\ &[(\vec{p}_1)^2 + (\vec{p}_3)^2 + \vec{p}^2][(\vec{p}_2)^2 + (\vec{p}_4)^2 + \vec{p}^2] = \\ &\quad - 4\lambda^2 i \frac{(t + 2m^2)^2}{t - m^2} \delta^{(4)}(\vec{P} - \vec{K}) \end{aligned} \quad (55)$$

$$\begin{aligned} \mathfrak{V}[6] &= (-2i\lambda)^2 i \int d^4 p \frac{\delta^{(4)}(\vec{p}_1 - \vec{p}_4 - \vec{p}) \delta^{(4)}(\vec{p} + \vec{p}_2 - \vec{p}_3)}{\vec{p}^2 - m^2 + i\epsilon} \\ &[(\vec{p}_1)^2 + (\vec{p}_4)^2 + \vec{p}^2][(\vec{p}_2)^2 + (\vec{p}_3)^2 + \vec{p}^2] = \\ &\quad - 4\lambda^2 i \frac{(u + 2m^2)^2}{u - m^2} \delta^{(4)}(\vec{P} - \vec{K}) \end{aligned} \quad (56)$$

FIG. 13. $\mathfrak{F}[7]$ FIG. 16. $\mathfrak{F}[10]$ FIG. 14. $\mathfrak{F}[8]$ FIG. 17. $\mathfrak{F}[11]$

$$\begin{aligned}\mathfrak{F}[7] &= (-2i\lambda)(6i\lambda m^2)i \int d^4p \frac{\delta^{(4)}(\vec{P} - \vec{p})\delta^{(4)}(\vec{p} - \vec{K})}{\vec{p}^2 - m^2 + i\epsilon} \\ [(\vec{p}_1)^2 + (\vec{p}_2)^2 + \vec{p}^2] &= 12\lambda^2 m^2 i \frac{s + 2m^2}{s - m^2} \delta^{(4)}(\vec{P} - \vec{K})\end{aligned}\quad (57)$$

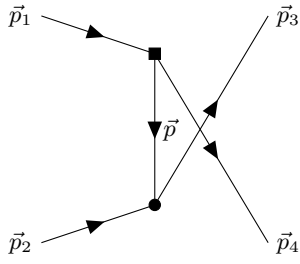
$$\begin{aligned}\mathfrak{F}[8] &= (-2i\lambda)(6i\lambda m^2)i \int d^4p \frac{\delta^{(4)}(\vec{P} - \vec{p})\delta^{(4)}(\vec{p} - \vec{K})}{\vec{p}^2 - m^2 + i\epsilon} \\ [(\vec{p}_3)^2 + (\vec{p}_4)^2 + \vec{p}^2] &= 12\lambda^2 m^2 i \frac{s + 2m^2}{s - m^2} \delta^{(4)}(\vec{P} - \vec{K})\end{aligned}\quad (58)$$

FIG. 15. $\mathfrak{F}[9]$

$$\begin{aligned}\mathfrak{F}[9] &= (-2i\lambda)(6i\lambda m^2)i \int d^4p \frac{\delta^{(4)}(\vec{p}_1 - \vec{p}_3 - \vec{p})\delta^{(4)}(\vec{p} + \vec{p}_2 - \vec{p}_4)}{\vec{p}^2 - m^2 + i\epsilon} [\vec{p}^2 + (\vec{p}_2)^2 + (\vec{p}_4)^2] \\ &= 12\lambda^2 m^2 i \frac{t + 2m^2}{t - m^2} \delta^{(4)}(\vec{P} - \vec{K})\end{aligned}\quad (59)$$

$$\begin{aligned}\mathfrak{F}[10] &= (-2i\lambda)(6i\lambda m^2)i \int d^4p \frac{\delta^{(4)}(\vec{p}_1 - \vec{p}_3 - \vec{p})\delta^{(4)}(\vec{p} + \vec{p}_2 - \vec{p}_4)}{\vec{p}^2 - m^2 + i\epsilon} [\vec{p}^2 + (\vec{p}_1)^2 + (\vec{p}_3)^2] \\ &= 12\lambda^2 m^2 i \frac{t + 2m^2}{t - m^2} \delta^{(4)}(\vec{P} - \vec{K})\end{aligned}\quad (60)$$

$$\begin{aligned}\mathfrak{F}[11] &= (-2i\lambda)(6i\lambda m^2)i \int d^4p \frac{\delta^{(4)}(\vec{p}_1 - \vec{p}_4 - \vec{p})\delta^{(4)}(\vec{p} + \vec{p}_2 - \vec{p}_3)}{\vec{p}^2 - m^2 + i\epsilon} [\vec{p}^2 + (\vec{p}_2)^2 + (\vec{p}_3)^2] \\ &= 12\lambda^2 m^2 i \frac{u + 2m^2}{u - m^2} \delta^{(4)}(\vec{P} - \vec{K})\end{aligned}\quad (61)$$

FIG. 18. $\mathfrak{F}[12]$

$$\begin{aligned}
 \mathfrak{F}[12] &= (-2i\lambda)(6i\lambda m^2)i \\
 &\int d^4p \frac{\delta^{(4)}(\vec{p}_1 - \vec{p}_4 - \vec{p})\delta^{(4)}(\vec{p} + \vec{p}_2 - \vec{p}_3)}{\vec{p}^2 - m^2 + i\epsilon} [\vec{p}^2 + (\vec{p}_1)^2 + (\vec{p}_4)^2] \\
 &= 12\lambda^2 m^2 i \frac{u + 2m^2}{u - m^2} \delta^{(4)}(\vec{P} - \vec{K})
 \end{aligned} \tag{62}$$

$$\begin{aligned}
 \mathcal{M} &= 4\lambda^2 m^2 - (6\lambda m^2)^2 \left(\frac{1}{s - m^2} + \frac{1}{t - m^2} + \frac{1}{u - m^2} \right) \\
 &- 4\lambda^2 \left[\frac{(s + 2m^2)^2}{s - m^2} + \frac{(t + 2m^2)^2}{t - m^2} + \frac{(u + 2m^2)^2}{u - m^2} \right] \\
 &+ 24\lambda^2 m^2 \left(\frac{s + 2m^2}{s - m^2} + \frac{t + 2m^2}{t - m^2} + \frac{u + 2m^2}{u - m^2} \right)
 \end{aligned} \tag{63}$$

$$\begin{aligned}
 \mathcal{M} &\propto m^2 - 9m^4 \left(\frac{1}{s - m^2} + \frac{1}{t - m^2} + \frac{1}{u - m^2} \right) \\
 &- \left[\frac{(s + 2m^2)^2}{s - m^2} + \frac{(t + 2m^2)^2}{t - m^2} + \frac{(u + 2m^2)^2}{u - m^2} \right] \\
 &+ 6m^2 \left(\frac{s + 2m^2}{s - m^2} + \frac{t + 2m^2}{t - m^2} + \frac{u + 2m^2}{u - m^2} \right)
 \end{aligned} \tag{64}$$

Using Mathematica I reduce this to

$$\mathcal{M} \propto 4m^2 - s - t - u. \tag{65}$$

$$\begin{aligned}
 s + t + u &= (\vec{p}_1 + \vec{p}_2)^2 + (\vec{p}_1 - \vec{p}_3)^2 + (\vec{p}_1 - \vec{p}_4)^2 = \\
 &6m^2 + 2\vec{p}_1 \cdot (\vec{p}_2 - \vec{p}_3 - \vec{p}_4)
 \end{aligned} \tag{66}$$

Conservation of momentum implies $\vec{p}_2 - \vec{p}_3 - \vec{p}_4 = -\vec{p}_1$, so

$$s + t + u = 6m^2 - 2\vec{p}_1 \cdot \vec{p}_1 = 6m^2 - 2m^2 = 4m^2. \tag{67}$$

$$s + t + u = 4m^2 \tag{68}$$

$$\mathcal{M} \propto 4m^2 - s - t - u = 4m^2 - 4m^2 = 0 \implies \boxed{\mathcal{M} = 0} \tag{69}$$

QED